

Low Contrast Image

a) Interpret the scenario & explain the cause of low contrast

A low contrast image is one in which the difference between the darkest & the brightest pixel intensities is very small. As a result objects & the background appear almost the same, making the image look dull & faded.

a) Causes:

- Poor lighting conditions
- Incorrect camera exposure
- Limited dynamic range of the camera sensor
- Fog, haze or smoke during image capture

b) Identify all key issues affecting image visibility

- Low overall image quality
- Reduced edge visibility
- Difficult object detection & segmentation
- Low overall image quality

c) Apply an appropriate enhancement technique.

Technique: Histogram Equalization

- 1) Calculate histogram pixel intensities.
- 2) Compute cumulative distribution function (CDF)
- 3) Redistribute pixel values across entire intensity range.
- 4) Generate enhanced image with improved contrast.

- d) Justify your choice with logical reasoning
- It increases the contrast automatically.
 - It spreads pixel intensities over the full grayscale range.
 - Hidden details become more visible.
 - It improves recognition without changing image content.

e) Clear & structured Explanation

Aspect	Explanation
Problem	Low contrast image
Cause	Small intensity difference
Issues	Poor visibility, hidden details, weak edges
Technique	Histogram equalization
Result	Better contrast & improved visibility.

2) Uneven Illumination

- a) Interpret the scenario & explain impact of even illumination
- Uneven illumination occurs when different parts of an image receive different amounts of light

Impact:

- Bright regions lose details due to overexposure.
- Dark regions hide important information.
- Object detection becomes unreliable
- Image appears inconsistent.

b) Identify key factors causing intensity variation

- Non-uniform lighting source.

- Uneven brightness
- hidden details
- Poor image quality

c) Enhancement Technique

CLAHE [Adaptive Histogram Equalization]

d) Enhance Justification:

- It improves brightness in each area & balance lighting

e) Result

- The image becomes bright sharper & balance light clear with uniform brightness

3) Salt - and - Pepper Noise

a) Cause:

The image contains random black & white dots because of noise

b) Issues:

- Poor readability
- Loss of image quality
- Text & objects become unclear

c) Technique

- It removes salt & pepper while keeping image edges clear.

d) Justification

- Median filter

e) Result :

The image becomes clear, clean & easy to read.

4) Blurred Edges

a) Cause

- The image becomes too smooth after noise removal, so the edges become blurred.

b) Issues :

- Loss of edge details
- Objects are not clear
- Reduced image sharpness

c) Technique :

Sharpening filter (High Pass filter)

d) Justification :

It enhances edges & make the image sharper.

e) Result :

The image has clear & better quality.

5) High Frequency Noise

a) Cause :

The satellite images contain fine random noise that affects visibility.

b) Issues :

- Poor image quality
- Difficult to identify features

c) Techniques : Low Pass filter.

d) Justification

It removes high frequency noise.

6) Object Boundary Detection:

a) Cause

- The boundaries between tissues must be identified for accurate medical ^{analysis}

b) Issues

- Boundaries are difficult to detect.

- Object may overlap

c) Techniques

Canny edge detection

d) Justification

It detects edges accurately while reducing noise

e) Result:

The tissue boundaries becomes clear & easy to identify

7) Simple Segmentation

a) Cause

- The object & background have different intensity values, so they can be separated easily.

b) Features

- Different brightness

- clear object boundary

c) Technique

Thresholding

d) Justification → It separates the objects from background using

a thresholding value.

Result:

The object is clearly separated from the background.

8) Complex Segmentation

a) Causes

The image has many regions with similar intensity values, making separation difficult.

b) Issues

- Intensity overlap
- Unclear boundaries

c) Technique

Water shed Thro Segmentation.

d) Justification

It separate touching & overlapping regions effectively.

e) Result

Different regions are accurate segmented.

9) Broken edges:

a) Cause:

Edge detection between Procedure gaps because of noise or over

b) Issues:

- Dis continuous boundaries
- Missing edge information

c) Technique : Morphological closing

d) Justification

It connects broken edges & fill small gaps.

e) Result :

The object boundaries becomes continuous & clear.

10) Combined Requirement :

a) Cause

The industrial image with continuous noise but important edges must be preserved to detect.

b) Issues

- Noise reduces image quality
- Blurring may remove important edges

c) Technique :

- 1) Median Filter
- 2) Canny Edge detection

d) Justification :

The median filter removes noise without affecting edges.

e) Result :

So, the Edges remain clear